

Dinghy / GPS-only SSA — prior art, and how to get TWD out of a track

Research behind the decision to adapt SSA for Olympic dinghy and keelboat classes that carry a **GPS tracker and nothing else**.

Three questions, answered in order:

- 1. Who already does this, how well, and what is left unserved? (§1–§4)
- 2. How do you derive TWD from a bare track, and how accurately? (§5–§9)
- 3. Should TWD and TWS come from *different* sources, as Njord does? (§11–§13)

Sources at the bottom. Several vendor pages are marketing copy with no technical detail; where a finding rests on a search summary rather than a primary page it is marked (*unverified*). Two primary sources refused the build environment (Wiley, the SAP wiki `concepts/` namespace — 401/403) and are marked as such.

1. Why this is a different product, not a setting

SSA today assumes the Northstar 76: an Expedition log with `twā`, `tws`, `twd`, `bsp`, `awa`, `heel`, rudder, forestay load, lidar sail shape, plus a KND event file whose `<phase>` elements pre-segment the day into 30 s steady-state chunks. Look at what the core modules actually consume:

module	depends on
<code>phaseStats.ts</code>	event-file <code><phase></code> list, <code>twā</code> sign for tack, polar
<code>manoeuvres.ts</code>	TWA sign flips, <code>bsp</code> , heading, <code>tws</code>
<code>startAnalysis.ts</code>	<code>twd</code> at the gun (line bias, <code>twdAtGun</code>)
<code>buildPhases.ts</code> / <code>phasePlot.ts</code>	mode from <code><sailingmode></code>

A dinghy tracker gives you `lat`, `lon`, `sog`, `cog` — and, on a Vakaros or Sailmon, `hdg`, heel and pitch. **Every single one of those modules' inputs is missing**. There is no polar, no event file, no phases, no sails to name.

The structurally right move is therefore *not* to fork the analysis. It is to insert a **synthesis layer** that manufactures the missing channels — `twd`, `tws`, `twā`, and a phase list — from the track, writes them into the existing `LogRow` shape (`{ utc } & Record<string, number|null>`, already open), and lets everything downstream run unchanged. TWD is the linchpin: get it, and `twā` follows, and with `twā` the existing mode classification, manoeuvre detection, VMG, start-line bias and phase stats all come back to life.

That is why the TWD requirement is the whole project, not a feature of it.

2. The landscape

2a. GPS-first analytics for dinghies — the direct competitors

tool	origin	data in	wind from track?	video	notable
kTool	HU, web	GPX, instrument logs, wind files, video	yes, stated as a headline feature	multi-angle sync	also has a Sail Shape Analyzer from photos — overlaps SailScan

tool	origin	data in	wind from track?	video	notable
SailViewer (IB-Sailing)	web/app	"GPS logs from any device", incl. Android phones	not stated	not stated	"used by Olympic teams around the world"; 2-month free trial
Vantage	iOS/Android/macOS	Vakaros, Sailmon, Garmin, Polar, Suunto, GPX, own GPS	yes — "true wind from GPS", ML-based (<i>unverified</i>)	telemetry overlay	Free / €-equiv \$119.99 yr PRO / \$179.99 yr Coach; 80+ classes
Kinetix AI	IL (Tel Aviv, f. 2020)	GPS tracker + GoPro	not stated	auto-cuts key moments into clips	built during an Olympic 49er campaign; from ~\$420/yr
SailSync	web	phone, Garmin, Vakaros, Strava, Velocitek, Sailmon	"wind & speed analysis"	video + transcribed audio	"FLO" AI assistant; \$20.99–62.99/mo coach tiers; 3D replay
OutSail.pro	EU, web	GPX + native Vakaros/Sailmon/Velocitek/Sailteck/Expedition	"wind shift tracking"	replay sync	free tier; €15/mo, €60/mo pro, €3,000/yr club; A-Cat beta 2026
RaceAnalyser (Kinetic Labs)	desktop + web	Vakaros RaceSense live + .vtx	yes — "derived from fleet tacks and gybes, no instruments needed"	no	PDF debriefs, "Pro Regatta Coach" NL Q&A; beta expires 30 Oct 2026
ChartedSails	web	GPX, Velocitek, Vakaros, GoPro, Garmin/Apple watch	partly — detects from track shape, but the documented workflow is <i>manual</i> : rotate the compass until wind is up	GoPro GPS extraction	seeds a baseline from fetched weather data
Fastrrr	HU	own sensors + Vakaros	not stated	—	Dinghy Edition / CoachBoat Pro / DataBay

tool	origin	data in	wind from track?	video	notable
Tactiqs	iOS	phone sensors + NMEA	tracks point of sail + shifts	—	52 metrics, AR sunglasses display

2b. Fleet/event tracking platforms (wind from the fleet)

tool	wind method	claimed accuracy
RaceQs	met data as prior, then a rule-based refinement on tacking/gybing angles and course shifts while close-hauled	3–5° single boat; ~1° with 5+ boats
SAP Sailing Analytics / Eclipse Azimuth	maneuver-based wind estimation with trained ML regressors (duration-based and distance-based, on TWD-delta measures); open-sourced Apache 2.0, Oct 2025	not published
TackTracker	"automatically estimates wind direction when analysing races", used to place sails, classify legs, compute VMG	not published
TracTrac	position/SOG/COG feed; wind comes from sensors, not the tracks	—
Kattack, SailRacer, GeoRacing	replay-grade	—

RaceQs' numbers are the most useful public benchmark in the whole field: 3–5° from one boat, ~1° from five. That is the bar. Note the second number — the fleet is not a nice-to-have, it is a 3–5× accuracy multiplier.

SAP going open source (Apache 2.0, 67k+ commits, Java/GWT) is the single biggest gift here. Their [com.sap.sailing.windestimation](#) bundle is the only production maneuver-based wind estimator whose source you can read. Worth a day of someone's time before writing a line of our own.

2c. Instrument-platform analytics (SSA's current neighbourhood)

Njord Analytics — "whatever instrument system or tracker your boat runs", auto-detects races and legs, builds a season database; **Njord Player** puts every camera angle and data channel on one timeline for the debrief, team views it in a browser while the coach builds it. Olympic dinghies listed alongside TP52/Maxi72/RC44. Pay-per-sailing-day or flat monthly. *This is the closest thing to SSA's debrief ambition, and it already claims Olympic coaches.*

Expedition / KND — where SSA lives now. Irrelevant to a boat with no instruments.

2d. Hardware the athletes actually have

device	logs	notes
Vakaros Atlas 2	25 Hz L1+L5 dual-band GNSS, 3-axis mag/gyro/accel	compass to 0.1°; pairs with a solar ultrasonic masthead wind sensor for real TWD/TWA
Vakaros Atlas Edge	same sensor suite, cut-down feature set, \$749	built for dinghies ; explicitly <i>no</i> wind-sensor support on a rotating mast
Sailmon MAX / Element	GPS + display + cloud	header/lift vs start-line wind direction; 3-minute wind plot; "wind from coaches or other close wind sources"
Velocitek ProStart / SpeedPuck	GPS	shift tracking : locks the mean heading, shows deviation as a bar, auto-resets on a detected tack — the single-tack half of the algorithm in §6c, in firmware since 2010
phone / Garmin / Apple Watch	1 Hz GPX	free; dunking risk
GoPro 5+	embedded GPS in the video	ChartedSails extracts it locally — one device for track <i>and</i> footage

The `.vkx` format is openly published by Vakaros on GitHub. Record 0x02 is position + SOG (m/s) + **COG (radians)** + altitude + an **orientation quaternion** — i.e. heel, pitch and heading are all recoverable. There are also records for declination (0x03), race timer (0x04), line position (0x05), a "shift angle" record (0x06: tack heading in degrees + speed in knots), wind (0x0A) and speed-through-water (0x0B) if sensors are attached. A .NET reference parser exists in the wild ([SCarlsen7757/Vakaros.Vkx.Parser.NET](#)). **Note: no licence file on the Vakaros repo** — check before vendoring anything.

That quaternion is useful — but **not for the wind**. See Part 4: measured against two real Atlas tracks, the magnetometer carries a 5–15° per-device bias while GPS COG agrees between boats to 0.2°. Heading is a *diagnostic* channel (manoeuvre detection, turn rate, low speed where COG is undefined), not the primary input to a wind estimate.

3. Class rules — the constraint nobody's marketing mentions

This shapes the product more than any feature decision.

- **ILCA**: electronic and digital compasses are **prohibited**. A compass or timing device "shall not be capable of displaying, delivering, transmitting, receiving, calculating, correlating or storing information about wind speed, wind direction, boat speed or boat position." A wrist-worn timing device with a non-GPS electronic compass is the exception. **A GPS logger is not class legal while racing.**
- **iQFOiL / Formula Kite**: personal GPS devices and recording equipment are **permitted when the NoR or SIs allow it**.
- **49er / 49erFX / Nacra 17 / 470**: governed by class rules plus the event's NoR/SI; events routinely *issue* trackers (the Nacra 17 Europeans required boats to carry event GPS units collected each morning).
- **RRS 41** (2025–2028) is about *outside help*, not onboard kit — equipment restrictions live in class rules. But 41(c) "information freely available to all boats" is what makes organiser-supplied fleet tracking legal.

Vakaros built **Class Compliance** profiles precisely for this: features (GPS input, distance-to-line, even magnetic heading) can be individually disabled, the device shows only what the class permits **while still logging the full session in the background**, and the log carries a locked record of which profile was active so a measurer can verify it. Several classes' measurers verify the profiles.

Product consequence, and it is a big one: for several Olympic classes the data model is *training-first*. Own-boat high-rate logging in training; racing data arrives later and from a different source — the event's tracker (TracTrac, SAP, RaceQs) at 1–5 Hz for the whole fleet. SSA needs **both ingestion paths**, and the fleet path is not an afterthought: §2b says it is where the wind accuracy comes from. The existing [docs/trac-trac-integration-research.md](#) is therefore not a side quest — it is on the critical path for the dinghy product.

4. What is actually unserved

The GPS-analytics space is crowded and converging fast: everyone does maneuver detection, VMG, replay, and increasingly video sync and an AI chat layer. Competing on "analyse my track" is competing on a commodity. Honest read:

- **kTool is the closest competitor to SSA's full shape** — wind from GPS, video sync, *and* sail shape from photos. It overlaps SailScan directly. Pricing page says "coming soon", so it is early. Worth watching closely.
- **Njord owns the credible Olympic-coach debrief story** on the instrument side.
- **RaceAnalyser** has the best-stated wind method and a principle worth stealing verbatim: "*honest data — marks unmeasured segments rather than inventing numbers.*" For a derived-TWD product this is not a nicety, it is the difference between a tool a coach trusts and one they don't.

What none of them appear to combine, and what SSA already has parts of:

1. **Weather depth** — SSA's forecast deck, 3D wind field, MOS, windweight, venue knowledge. The GPS tools all treat wind as a per-session scalar. For an Olympic campaign the venue model *is* the product.
2. **Media as a first-class citizen** — photos, video, drone, speed-team compilations and documents, shared across a squad. Kinetix does video; nobody does the day's whole media record.
3. **Campaign continuity** — SSA's campaign spine, session→event→season. The competitors are session tools.
4. **Squad-level training-group analysis** — 3–6 boats of the *same* team training together is the Olympic reality, and it is also (§5) where TWD accuracy jumps from 3–5° to ~1°. Nobody is selling that as the headline.

The defensible position is therefore: **the venue + the squad + the whole day's record**, with track analysis as table stakes. Not "another tracker app".

5. TWD from a track — framing the problem

Everything below assumes: position at 1–25 Hz, SOG, COG, optionally heading and heel, no wind sensor, no speed-through-water, no polar to start with.

Read Part 4 before implementing any of it — the method ladder below was written from desk research and has since been run against two real Atlas tracks. The main correction: prefer **COG over heading** throughout.

What you are really solving

The observable is velocity **over ground**:

$$V_{\text{ground}} = V_{\text{water}}(\text{TWA}, \text{TWS}) + C$$

C is the tidal/current vector. The sailor chooses TWA; the boat's polar maps (TWA, TWS) → speed through water. You want TWD, and TWD only enters through the *rotation* of V_{water} into the earth frame.

Three unknowns hide behind one observation: **what the wind is doing, what the water is doing, and what the sailor chose to do**. Every method below is a different way of breaking that three-way ambiguity.

Why this is harder than the forum answer

The standard answer — "average your two close-hauled headings" — fails in four specific ways, and each one is worth a named mitigation:

1. **It needs heading, and you have COG**. They differ by leeway plus current crab. On an ILCA upwind that is several degrees of leeway alone.
 2. **Cross-wind current rotates the bisector** (§6e — with a formula and a number). This is the dominant error and it is *systematic*, not noise.
 3. **TWD is not constant**. Oscillations of 5–10° on 3–8 minute periods are the entire tactical game. A session-mean TWD answers the wrong question.
 4. **Tacks are sparse**. One every 1–3 minutes upwind, none at all down a run. A tack-only method has terrible temporal resolution exactly where you need it.
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6. The method ladder

Build these in order. Each is useful on its own and each feeds the next.

L0 — Priors: model, marks, committee

Free, and they bound everything else.

- **Weather model.** SSA already fetches and stores this. ChartedSails does exactly this as a baseline and RaceQs seeds its algorithm from met data before refining. Gives TWD to maybe $\pm 15\text{--}20^\circ$ inshore — useless tactically, invaluable as a prior that resolves the 180° ambiguity and catches gross failures.
- **Course geometry.** A race committee sets the windward mark square. The start-line normal and the leg bearing are both TWD estimates to $\sim \pm 5\text{--}10^\circ$. Mark positions can be recovered from rounding patterns — RaceQs claims **~95% from one boat, ~99% from several**, plus a static-mark database. SSA would derive them from the track's turn clusters.
- **Coach boat.** The highest-value cheap upgrade in the whole project: an ultrasonic anemometer on the RIB (Calypso portable, or the Vakaros solar masthead unit on a pole) logging alongside its own GPS. Subtract the RIB's motion vector and you have *measured* TWD to feed in as truth. This is what the pro teams do, and it doubles as the validation set for §8.

L1 — Tack-pair bisector

The baseline. For each detected tack, take a steady-state window before and after (SSA's `manoeuvres.ts` already uses $-40\ldots-10$ s and $+17\ldots+23$ s windows for the N76 — same idea, retuned), and bisect.

Refinements that matter:

- Bisect **COG**, not heading (Part 4 — heading carries per-device compass bias; leeway is symmetric between tacks and cancels in the bisector).
- Use the **circular** mean — `startAnalysis.ts` already has `circularMean`.
- Weight by steadiness: reject windows containing a gust response, a wave set, a big heel excursion, or a second manoeuvre inside the settle time (SSA already has `shortHitch` for this).
- Reject any window whose speed is below a fraction of the running best for the conditions — a pinching or stalled boat lies about its angle.

Expected: **3–5°**, matching RaceQs' single-boat claim, and biased by current.

L2 — Steady-segment fitting, not just tack pairs

Better: segment the whole track into straight-line steady runs, classify each as upwind/reach/downwind, and fit a TWD that makes the population of upwind COGs bimodal and symmetric. This uses every second on the beat, not just the two windows either side of a tack, and it copes with legs where the boat tacked once.

This is where the **chicken-and-egg** bites: you need TWD to classify point of sail, and point of sail to estimate TWD. Solve it as **EM**: initialise from L0, classify, re-fit, re-classify, iterate. It converges quickly in practice because upwind COG distributions are sharply bimodal. `mkobetic/gpx` takes the same shape — segment into moving/turning/static, then analyse the moving segments — and notably *warns and skips* the analysis when the determination fails rather than guessing. Steal that behaviour.

L3 — Single-tack shift tracking (the high-rate channel)

L1/L2 give you **absolute** TWD at sparse times. Between tacks, hold the TWA estimate fixed and read COG deviation directly as a wind shift. This is exactly what the Velocitek SpeedPuck/ProStart shift tracker does in firmware — lock the mean heading, display deviation, auto-reset on a detected tack — and what Sailmon exposes as header/lift against the start-line wind direction.

The failure mode is that a COG change might be the *sailor*, not the wind. **Use SOG as the discriminator:**

COG	SOG	reading
up (toward the wind)	steady or up	lift
up	down	pinching — sailor, not wind
down	up	header , or footing for a gust
down	steady	header

Fuse: L1/L2 as sparse absolute anchors, L3 as the high-rate relative signal between them. That combination is what produces a usable TWD(t) trace at the resolution the tactical debrief actually needs.

L4 — Joint polar-constrained solve for wind and current

The one that actually fixes the systematic error. Unknowns: `TWD`, `TWS`, `C_x`, `C_y` — four. Each steady segment on a known tack contributes two equations (the two components of `V_ground`). If you assume the sailor sails the polar's target VMG angle, then `TWA* = f(TWS)` and the segment adds **no new unknowns**: two opposite-tack segments give four equations in four unknowns, exactly determined. Relax the target-angle assumption and let TWA float per segment with a prior, and each segment adds one unknown for two equations — so four or more segments, over-determined, least squares.

Downwind segments help disproportionately: gybe angles differ from tack angles, which breaks degeneracies that upwind-only data leaves.

The key structural insight: `TWD` varies on a 3–8 minute timescale; `C` varies on an hours timescale and is near-constant over a training session. So run it as a state-space model — a random walk with **high process noise on TWD, near-zero on C** — and pool the whole session. Current becomes strongly observable from the session as a whole, and once `C` is pinned down, *every individual tack pair yields a clean, unbiased TWD*.

This is, as far as I can tell from the public material, **not what any of the competitors do**. RaceQs says its VMG "incorporates current effects" without claiming to solve for the current vector; everyone else is silent. It is the one place where SSA could be genuinely better rather than equal.

Independent cross-check when heading is available: `COG - HDG` is leeway + current crab + **compass error**, and Part 4 shows how to separate them — the part that flips sign between tacks is leeway, the tack-independent part is compass offset or cross-wind current, and those two are algebraically degenerate. In non-tidal water the common mode is the compass; in tidal water this cross-check only works once the compass has been calibrated elsewhere.

L5 — Pool the fleet / the training group

RaceQs: $3-5^\circ \rightarrow \sim 1^\circ$ going from one boat to five. Three effects stack: averaging independent errors; different boats tacking at different times giving denser temporal sampling; and boats spread across the course letting you fit a *spatial* field rather than a single number.

For a national squad this is free — they already train 3–6 boats together. Pool every boat's L2 segments into one estimator, and for a regatta pull the event tracker's whole fleet.

The spatial step (kriging / Gaussian-process regression over the course, as used for aircraft-derived wind fields) is the natural extension and answers the question coaches actually ask — *which side had more?* — rather than just *what was the mean wind?* This is a phase-2 item but it is where the fleet data really pays.

L6 — Learned models

Two references:

- **SAP** trains regressors for maneuver-based wind estimation, on both duration-based and distance-based features of TWD-delta measures (std dev, mean). Their training pipeline is two-stage, wants ≥ 100 GB and 24+ hours on 16 GB RAM. Open source.
- `neil-marcellini/ml-wind-estimator` — a PyTorch model for exactly this problem, stated goal "accurate wind direction estimate at each point, live VMG, and segmenting tracks without recording wind data." Small, but it is the hobbyist proof that the supervised framing works. (*README beyond the summary line was not retrievable; the architecture is in net.py.*)
- Adjacent literature: **MoWe** (Sun et al., *J. Field Robotics* 2025) — motion-observation wind estimation for sailing robots, combining a motion-analysis component with a data-driven one (*paper behind a 403; method summary from search results only*). LSTM-on-roll-and-pitch is the recurring architecture in the quadrotor analogue.

Do not start here. ML is the right layer once there is a labelled corpus — and the corpus comes from L0's coach-boat anemometer plus the squad's own sessions. Physics first, learning as the residual correction.

7. The current-bias formula

Worth writing down because it sets the whole error budget.

Take symmetric upwind TWA, speed through water V , and a current with component $c_{\perp}$ perpendicular to the wind axis. The bisector of the two ground tracks is rotated away from true TWD by:

$$\begin{aligned}\text{bias} &\approx (c_{\perp} \cdot \cos \text{TWA}) / V && \text{radians} \\ &\approx 57.3 \cdot c_{\perp} \cdot \cos \text{TWA} / V && \text{degrees}\end{aligned}$$

Numbers:

boat	V (kn)	TWA	$c_{\perp} = 0.5 \text{ kn}$	$c_{\perp} = 1.0 \text{ kn}$
49er upwind	6.0	42°	3.6°	7.2°
ILCA upwind	4.5	45°	4.5°	9.2°
470 upwind	5.0	43°	4.2°	8.5°

(Exact, from the two-tack geometry; the formula above is the small- c linearisation and is within 0.2° of exact out to 1 kn.)

Half a knot of cross-wind tide puts 3.5–4.5° of *systematic* error into TWD — comparable to the entire oscillation amplitude you are trying to measure. In Marseille, Weymouth, Hyères or the Solent this is not a rounding error, it is the answer.

Two further consequences:

- A current component **along** the wind axis does **not** rotate the bisector. It changes the apparent tack angle instead — so it corrupts your TWS and polar estimate, not your TWD. Useful: the two error modes are separable.
- Which means an L1 bisector is unbiased in TWD when the tide runs straight up or down the course, and worst when it runs across. Worth surfacing in the UI as a confidence modifier rather than hiding it.

8. Validation — do not ship this unvalidated

The one thing that will sink a derived-TWD feature with an Olympic coach is being confidently wrong once. Plan:

1. **Ground truth:** a coach-boat ultrasonic anemometer + GPS, motion-corrected. One RIB, one afternoon, and you have a labelled day.
2. **Cross-source:** SSA's own forecast/MOS stack as a sanity bound; the committee's recorded wind at a regatta; the event tracker's fleet-derived wind if TracTrac/SAP supply one.
3. **Self-consistency:** TWA distribution symmetry between tacks after the fit; L4's two independent routes to C agreeing; estimated TWS against the polar the same data implies.
4. **Report the band, always.** Per-segment confidence, and an explicit "*unmeasured*" state where the estimate fails. RaceAnalyser's "honest data" principle, and `mkobetic/gpx`'s warn-and-skip, both get this right. A derived number that does not carry its uncertainty is worse than no number in a debrief, because it gets argued about.

Target to beat, stated publicly by a competitor: **3–5° single boat, ~1° with a fleet of five**. If SSA's L4 current-corrected estimator does not beat 3–5° on a tidal venue, it has not earned its complexity.

9. What this implies for the SSA codebase

Architecture — synthesis layer, not a fork:

```

tracker file (.vtx / GPX / Sailmon CSV / TracTrac)
  ↓ new parsers alongside logParse / flatLogParse / csvLogParse
LogRow[] { utc, lat, lon, sog, cog, [hdg, heel, pitch] }
  ↓ NEW: src/lib/wind/segment.ts — steady/turn/static segmentation
  ↓ NEW: src/lib/wind/estimate.ts — L0→L4 ladder, returns twd/tws + confidence
LogRow[] + { twd, tws, twa, bsp- sog } ← existing channel names, unchanged
  ↓
phaseStats · manoeuvres · startAnalysis · phasePlot (unchanged)

```

Concrete consequences, in dependency order:

1. **Phases must be derived.** There is no KND event file. The steady-state segmenter from L2 is the phase builder — `buildPhases.ts` gains a track-derived source. This is a prerequisite for wind estimation, and wind estimation is a prerequisite for classifying the phases. Build it as the EM loop described in L2; do not try to do them separately.
2. **Polars must be learned, not loaded.** No dinghy class ships a polar. Bootstrap from published class target angles, then refine from the boat's own data once TWD is stable — another EM loop, and one that gets better across a season. `polarCalc` needs a "learned polar" provenance flag so nothing downstream mistakes an inferred polar for a measured one.
3. `bsp` becomes `sog`. Legitimate for a dinghy with no paddlewheel, but it means every existing "boat speed" metric silently becomes speed over ground and inherits the current. Needs to be explicit in the data model, not a coincidence of naming — this is exactly the class of trap CLAUDE.md exists to record.
4. `manoeuvres.ts` **mostly survives**. It already has a TWA-sign-flip fallback path for when the event file is absent. The KND-fitted windows and the `minBsp: 6 kn` floor are N76 numbers and need per-class retuning.
5. **Confidence has to reach the UI.** Every derived channel needs a provenance + uncertainty pair carried alongside it, and the charts need an "unmeasured" rendering. Retrofitting this later will be painful; the `LogRow` shape is open enough to carry it from the start.
6. **Fleet ingestion is on the critical path**, not optional (§3, §5-L5). The TracTrac research already done is the other half of this product.
7. `.vtx` **first**. It is the open format, the Atlas Edge is the dinghy device at \$749. The quaternion gives heading, heel and pitch — worth parsing for manoeuvre detection and for the per-device compass calibration of Part 4, though not as the wind input. GPX second as the universal fallback.

Before writing the estimator: read SAP's `com.sap.sailing.windestimation` bundle (Apache 2.0). It is the only production implementation of maneuver-based wind estimation with readable source, and a day spent there is cheaper than a week of rediscovery.

10. Open questions

- Does kTool's "wind direction from GPS track" handle current? Their sail-shape overlap with SailScan makes them the competitor to evaluate properly — sign up for the free tier and test it on a known tidal day.
- Can we get the SAP wiki `concepts/windestimation` page? It 401'd from here; the `howto/` page is public and describes only model *training*, not the method.
- What do the Dutch/other squads currently use, and what do they complain about? This whole doc is desk research — one hour with a squad coach beats all of it.
- Licence on `vakaros/vtx` — the repo has none. Ask before vendoring.
- Is per-boat wind estimation even the right unit for a squad, or should the primitive be a **course wind field** with per-boat observations feeding it? §5-L5 argues the latter; that is a bigger architectural commitment and worth deciding early rather than refactoring into.

Sources

Tools and platforms:

- [Njord Analytics](#) · [Sailmon data source page](#)
- [Vantage](#) · [pricing](#)
- [kTool](#)
- [SailViewer](#) / [IB-Sailing](#)
- [Kinetix AI FAQ](#) · [Sailing World: Kinetix, The Debrief Game-Changer](#)
- [SailSync](#)
- [OutSail.pro](#)
- [RaceAnalyser](#), [Kinetic Labs](#)
- [ChartedSails — 8 ways to record GPS tracks](#) · [ChartedSails](#)
- [Fastrrr](#)
- [Tactiqs](#)
- [TackTracker analysis](#)
- [SailZing — Sailboat Race Analysis](#)
- [Sailing World — Where Data Meets Development](#)

Wind estimation methods:

- [RaceQs — How RaceQs calculates Wind, VMG and Mark Locations](#)
- [SAP Sailing Analytics wiki — wind estimation model training](#)
- [SAP/sailing-analytics on GitHub](#) · [Sailing Analytics Goes Open Source](#) · [Eclipse Azimuth proposal](#)
- [neil-marcellini/ml-wind-estimator](#)
- [mkobetic/gpx](#)
- [MoWe: Motion Observation for Wind Estimation of Sailing Robots \(403 — summary only\)](#)
- [Wind velocity field estimation from aircraft derived data using Gaussian process regression](#)
- [Performance Analysis in Olympic Sailors of the Formula Kite Class Using GPS](#)

Hardware and formats:

- [Vakaros Atlas 2](#) · [Atlas Edge](#) · [Atlas 2 on Any Boat](#) · [What is Class Compliance?](#) · [Ultrasonic Wind Sensor](#)
- [vakaros/vkx — the VKX telemetry log format](#) · [SCarlsen7757/sail-sight](#)
- [Sailmon MAX](#) · [Header/Lift, Wind Trends and app updates for MAX](#)
- [Velocitek ProStart shift tracking](#) · [SpeedPuck](#)

Rules:

- [ILCA Class Rules](#)
 - [World Sailing — 2025-2028 Racing Rules of Sailing](#) · [RRS Part 4, other requirements when racing](#)
 - [World Sailing — 2026 Class Rules, iQFOiL](#) · [IKA Formula Kite](#)
-

Part 3 — Decoupled wind sources

Added after reviewing Njord's **Modify Wind Direction / Speed** dialog (screenshot, TP52 fleet, eight TracTrac boats, 11:00–16:00 GMT+0, races 10–12). Their user guide is public and confirms what the dialog shows.

11. What Njord actually ships

Two **independent** dropdowns, and the option lists are not the same:

	Wind Direction Source	Wind Speed Source
inferred from the track	Inferred from typical TWA	— <i>(nothing)</i>
model	Weather Data (OpenWeather Time Machine)	Weather Data (OpenWeather Time Machine)
manual, fixed	Constant Value	Constant Value
manual, time-varying	Interpolated over Time	Interpolated over Time

Their description of the inference method, verbatim from the dialog:

Determines typical upwind and downwind TWA from the best maneuvers; then assumes that straight line sailing is done at this TWA and infers TWD from the boat's course. This method is able to detect wind shifts between maneuvers.

That is **exactly the L1+L3 pair from §6**, shipped: calibrate a typical TWA off the good manoeuvres, then read the course between manoeuvres as shift. The `Initial TWD: Auto` field is the L0 seed, and their guide recommends seeding it with **the first start-line bearing** — the course-geometry prior from §6-L0.

What their docs concede about when it fails, all of it consistent with §6:

- works best with "clean straight upwind/downwind sailing, clear tacks and gybes, and consistent wind"; poor in light, shifty, or reaching-dominated days
- "performs better when multiple boats' datasets are loaded together"** — the fleet multiplier again, now confirmed by a second vendor
- defining the full course (all marks) is more robust than TWA-based leg detection alone** — marks as hard geometric constraints
- when it drifts mid-day: shorten the time range, seed the initial TWD, or run **separate inferences per segment**
- a coach-boat wind file can supply TWD/TWS directly

The asymmetry is the whole point

TWD has an inference method. TWS has none. That is not an oversight, and it is not a roadmap gap — it follows from observability:

	observable from a bare track?	available from a model?
TWD	strongly — it is a pure <i>geometry</i> problem. Angles are what a track is made of.	badly at the scale that matters. Models get the synoptic mean; the 5–10° oscillation on a 3–8 min period is entirely sub-grid.
TWS	weakly — needs a polar to convert speed into wind, and you have SOG not BSP.	reasonably . Magnitude is the thing NWP does comparatively well, albeit smoothed.

So each channel is taken from whichever source is actually good at it. Once you see it that way the decoupling is not a clever UI trick, it is the only defensible design — and it generalises past two channels (§13).

The correct name for what a track gives you

Worth fixing the vocabulary, because it makes §7 exact rather than a caveat. Following Burch's distinction (and Expedition's, and B&G's):

- **True wind** is referenced to **the water**, computed from **STW / CTW**. It is what the sailor trims to, and what polars are tabulated against.
- **Ground wind** is referenced to **the ground**, computed from **SOG / COG**. It is what forecasts are issued in.
- They differ by exactly the current vector. Burch's Gulf Stream worked example: 3 kn of current produced a **4° direction and 3 kn speed difference** between the two.

A GPS-only track therefore yields **ground wind, natively and only**. Everything in §6-L1 through L3 estimates **GWD, not TWD**. §6-L4's joint solve for the current vector is not an accuracy refinement bolted on the side — *it is* the GWD → TWD conversion, and without it every "TWD" the tool prints is mislabelled.

This also explains the §7 error budget in one line: on a tidal venue, a tool that infers wind from a track and calls the answer TWD is reporting ground wind with a true-wind label, and the gap is the current. **Nobody in §2 appears to name this distinction in their GPS-only path**. Getting the two labels right — and offering both — would be a small, cheap, genuinely differentiating piece of rigour.

12. The same idea, two other ways

SAP goes further than a dropdown. Wind is modelled as **multiple named source tracks plus a combination**: release notes reference source types `EXPEDITION`, `WEB`, `WINDFINDER`, `RACECOMMITTEE`, their maneuver-based estimator, and GRIB import that "turn[s] each position into a separate wind source, leading to better visual interpolation" — all reducible to a **COMBINED** wind track, which is exported alongside the individual ones. Crucially, **per-race source de-selection is persisted** and "survive[s] re-loading of the race as well as re-starting or upgrading a server."

Three lessons there, all cheap to copy:

1. A source is a **time series**, not a setting. Several coexist.
2. The combination is **derived and inspectable**, and the inputs stay visible next to it. (The screenshot shows Njord doing the lightweight version of this: the grey existing trace under the orange preview.)
3. The curation decision — *this source is rubbish for this race* — is **data, and it persists**. It is not a transient UI state.

Expedition takes the raw sensors independently (CTW, STW, COG, SOG) and computes TWS/TWD/**GWD** itself rather than trusting the instrument's own true wind, with TWD offset correction on top. Its canonical calibration check is "compare TWD when sailing at the same angle on port and starboard" — i.e. the §6-L1 bisector, used as a *calibration residual* rather than as an estimator. Same geometry, pointed the other way.

13. What this means for SSA

13a. SSA's model baseline is already better than Njord's

Njord's "Weather Data" is **OpenWeather Time Machine**: hourly, global, underlying model not publicly stated. You can see the consequence in the screenshot — the TWS trace is a 6 → 9 → 8 → 10 kn staircase of linearly interpolated hourly points, with every puff and lull gone, sitting under a TWD trace full of real structure.

SSA already fetches **Open-Meteo's Historical Forecast API**: hourly as standard but **15-minutely across Central Europe and North America**, from **ICON-D2 at 2 km**, UKMO UK 2 km and HARMONIE AROME, with gusts as a separate variable — plus SSA's own MOS bias correction and windweight layer on top.

That is a materially better TWS source than the market leader ships, **and it is already built**. It is the least effortful differentiator in this whole document: plug the existing weather stack in as a named wind source. (Watch the per-location billing — a wind track along a boat's path is a sequence of positions, and Open-Meteo bills per location; snap to a coarse grid and interpolate.)

13b. Prefer additive decomposition over source-picking

A dropdown makes the sources mutually exclusive. They are not — they are good at different *frequencies*. Better:

```

TWD(t) = model_baseline(t)      ← synoptic trend, gradient, sea-breeze veer
      + track_deviation(t)      ← the oscillation, from L1/L3
      - current_correction      ← the GWD→TWD step, from L4

TWS(t) = model_baseline(t)      ← including gusts, at 15 min / 2 km
      + polar_residual(t)       ← optional, once a learned polar exists

```

The model contributes what it is good at (slow, large-scale, unbiased-ish); the track contributes what it is good at (fast, local, relative). This also fixes the mid-day-drift problem Njord tells users to work around by manually chopping the time range into segments — an inferred-only TWD has no anchor and wanders, while an additive one is pinned to the model's trend by construction.

Keep the dropdown as the manual override. Make the additive fusion the default.

13c. Can TWS be inferred at all? Partly — and it doesn't matter much yet

Nobody does it. Ranked by how much signal is actually there for a dinghy:

	signal	verdict
S1	model baseline	use it — the honest default
S2	mode transitions : planing on/off, foiling take-off and touch-down	strongest single-boat signal . A sharp, class-specific threshold (iQFOiL, Nacra 17, 49er downwind). Nobody uses it.
S3	tack / gybe angle vs breeze	real, class-specific, non-monotonic at the extremes
S4	polar inversion from SOG	needs a learned polar <i>and</i> the current — circular until L4 lands
S5	heel / trapeze-vs-hiking mode from the Vakaros quaternion	crude but free once you parse the quaternion
S6	fleet mode fraction — what share of the fleet is planing	a soft fleet-wide anemometer; only available on the fleet path

Verdict: **S1 now, S2 next, the rest later**. And be clear about the priority — a TWS error mostly corrupts target-speed comparison, while a TWD error corrupts TWA, VMG, mode classification, lift/header and start-line bias *directly*. The engineering effort belongs in TWD and the current vector, not in squeezing TWS out of a track.

13d. Architecture: a wind-source registry, not a pair of settings

Concretely, for `src/lib/wind/`:

```

type WindSourceKind =
  | 'model'          // SSA weather stack: Open-Meteo / MOS / windweight
  | 'inferred'       // L1-L3 from the track — produces GROUND wind
  | 'solved'         // L4 joint wind+current — produces TRUE wind + current
  | 'fleet'          // L5 pooled across the training group / event tracker
  | 'sensor'         // coach-boat anemometer, masthead unit, committee log
  | 'manual'         // constant or interpolated, the human override

interface WindTrack {
  kind: WindSourceKind
  reference: 'ground' | 'true' // NEVER inferred — always declared
  twd?: Series; tws?: Series   // a source may supply only ONE channel
  confidence: Series           // per-sample, not per-track
  provenance: { model?: string; boats?: string[]; window?: [number, number] }
  enabled: boolean             // persisted per session/race, SAP-style
}

```

Then a `combine(tracks) → WindTrack` that does §13b, and the existing `LogRow` gets `twd / tws` written from the combined result with provenance carried alongside.

Four design rules falling out of the research:

1. **A source supplies channels, not "wind".** Njord's two dropdowns are the minimum viable version of this; make it the type.
2. **reference is declared, never assumed.** This is the ground-vs-true trap in §11, and it is exactly the class of thing CLAUDE.md exists to record — like the local-wall-time clocks, it silently produces plausible wrong answers.
3. **Enabled/disabled per source per session, persisted.** SAP's lesson.
4. **Recompute derived channels automatically.** Njord's FAQ makes users manually *drop* TWA/TWS/VMG, apply a formula, then add a "Derived Metrics" step to recalculate — and warns that "applying a formula twice applies it cumulatively," with a Reset-to-uploaded-files escape hatch. That is an invalidation bug surfaced as a user-facing procedure. SSA should track the dependency graph (`twd → twa → vmg → phaseStats → manoeuvres`) and recompute on change. Their FAQ also notes polars are "applied at upload time only" — same failure, same fix.

13e. The batch affordance is worth copying outright

The screenshot applies one wind solution to **eight boats over a five-hour window spanning three races** in a single action. For a squad training day — 6 boats, one wind — that is precisely the right unit of work, and it is *also* what §5-L5 needs: the fleet is the estimator's input, so the fleet should be the estimator's scope. Do not build this per-boat and generalise later.

14. Revised open questions

- Does Njord's inference return **ground wind labelled as TWD**? Their docs don't say, and the dialog doesn't. If so it is a real, checkable accuracy gap on tidal venues — and the first thing to test on a Solent or Hyères day.
- Which model does OpenWeather Time Machine actually use for wind? Not published as far as I can find. If it is ERA5-family at ~25 km, the gap to ICON-D2 at 2 km is larger than §13a implies.
- SAP's `COMBINED` wind track: what is the actual combination rule — nearest, weighted, confidence-based? The source is Apache 2.0, so this is readable rather than guessable, and it is the one published implementation of §13b.
- Does SSA's Open-Meteo budget survive per-boat wind tracks? (See the per-location billing note in the existing Open-Meteo memory.) Grid-snap first, measure second.

Sources added for Part 3

- [Njord user guide — Adding Wind Data](#) · [FAQ](#) · [Overview](#) · [Data Sources](#)
 - [Njord — Using Inferred True Wind Direction \(video\)](#) · [Adding Wind Data from Any Source \(video\)](#) (*not retrievable from the build environment*)
 - [SAP Sailing Analytics — Admin Console release notes](#) (wind source types, COMBINED track, persisted per-race de-selection, GRIB-per-position)
 - [David Burch — True Wind and Ground Wind, and Why We Need Both](#) · [B&G glossary](#)
 - [Expedition manual](#) (computes TWS/TWD/GWD from raw CTW/STW/COG/SOG; TWD offset correction; port/starboard calibration)
 - [Open-Meteo Historical Forecast API](#) (15-minutely, ICON-D2 2 km, HARMONIE AROME)
 - [OpenWeather One Call 3.0 / Time Machine](#) · [History API by timestamp](#)
 - [Observation-driven correction of numerical weather prediction for marine winds](#) (assimilating point observations to correct model wind — the formal version of §13b)
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Part 4 — Field validation against two real Atlas tracks

Parts 1–3 were desk research. This part is what changed after running the method ladder against **two Vakaros Atlas CSV exports from the same session** — Bay of Palma, 8 Feb 2026, 11:56–15:09 local (+0100), two boats training together.

The session turned out to be close to ideal for this: median separation **36 m**, 87 % of the time within 100 m, three hours. Two boats that close share one wind, so **any disagreement between them is instrument or method error, not weather**. That is the strongest validation signal available without an anemometer, and it costs nothing.

15. What the Atlas CSV export actually contains

```
timestamp,latitude,longitude,sog_kts,cog,hdg_true,heel,trim
2026-02-08T11:56:40.049+0100,39.5310307,2.5704827,0.400,266.1,245.8,-1.8,7.7
```

Eight channels, including true heading, heel and pitch. Findings that matter for the parser:

- **Timestamps are ISO 8601 with an explicit UTC offset.** Unambiguous. This is the *one* clock in SSA's world without the local-wall-time-labelled-UTC trap in CLAUDE.md — record it as the exception, and take the session date from the first timestamp.
- **CRLF line endings.** A naive `split(',')` on the header produces a final column named `trim\r`, so `trim` silently reads as undefined on every row. Cost: one wrong conclusion ("the pitch channel is garbage") before it was spotted. Strip `\r` before parsing.
- **No device identity of any kind** — no serial, no boat name, no session header. Confirms that the `.vtx` finding in §3b of the multi-boat doc extends to the CSV export. Identity lives only in the filename.
- **Filenames are user-typed and must never be parsed.** The two files here were `Miss Behavior 2 2-8-2026.csv` and `Torvar's second 08-02-2026.csv` — the same day written `M-D-YYYY` and `DD-MM-YYYY`, one naming a *boat* and the other naming a *session*, plus a curly apostrophe and a double space.
- **Logging rate is per-device and differs within a session:** 2 Hz (22,812 rows) and 10 Hz (104,617 rows) on the same afternoon. Resampling to a common grid is mandatory, not an optimisation.
- Gaps happen: six gaps > 5 s on one device (longest 22 s), none on the other.
- **Attitude plausibility varies enormously between devices.** One logged heel spanning $-173^\circ \dots +133^\circ$ and pitch $-88^\circ \dots +82^\circ$; the other $-42^\circ \dots +48^\circ$ and $-8^\circ \dots +25^\circ$. Attitude needs outlier rejection before use, and a badly mounted device is visible in the data.

16. The correction: COG beats heading, and it is not close

Running the L2 symmetry estimator on each boat independently, whole session:

channel used	boat A	boat B	disagreement
<code>hdg_true</code> (magnetometer)	259.8°	249.7°	10.2°
<code>cog</code> (GPS Doppler)	265.1°	264.9°	0.2°

Two boats 36 m apart cannot experience a 10° difference in wind direction. The disagreement is **per-device magnetometer bias**.

This inverts the recommendation in §6-L1 and §2d. The reasoning that led the desk research astray was that heading is closer to *through-water* direction, so it should be the cleaner wind input. What that missed:

Leeway is symmetric between tacks and therefore cancels in any bisector or symmetry fit. It widens the apparent tack angle — it does not bias the estimate of where the wind is coming from. Compass error does not cancel, and on these devices it is an order of magnitude larger than the thing heading was supposed to fix.

COG is the primary wind channel, always. Heading is a diagnostic channel — manoeuvre detection, rate of turn, and low speed where GPS COG becomes undefined — not an input to the wind estimate.

17. Separating compass error from leeway (and why "non-tidal" matters)

$\text{crab} = \text{COG} - \text{HDG}$ decomposes on opposite upwind tacks:

- the part that **flips sign** between tacks is **leeway**
- the **tack-independent** part is a **compass offset**

Measured, at the device-independent TWD of 265°:

boat	crab, tack 1	crab, tack 2	compass offset δ	leeway
Miss Behavior 2	+6.9° (heel 20°)	+3.4° (heel -17°)	-5.1°	1.8°
Torvar 2nd	+19.8° (heel 19°)	+9.3° (heel -19°)	-14.5°	5.3°

Leeway of 2–5° upwind is physically sensible. And the decomposition **cross-validates**: these offsets predict a heading-derived disagreement of **9.4°** between the two boats; the independent symmetry fits measured **10.2°**. Two unrelated routes, one answer.

(Sample: ~20 steady upwind segments per boat per tack, no standard errors computed — treat δ as ± 2 –3°.)

The degeneracy

A cross-wind current produces exactly the same common-mode crab signature as a compass offset. They are algebraically indistinguishable in this decomposition. Palma in February has negligible tide, so the common mode is attributable to the compass. On a tidal venue the same -14.5° could be half a knot of cross-tide.

Hence the operational rule:

Calibrate devices on non-tidal training days, then carry the per-device offsets to tidal venues — where a calibrated compass becomes valuable again, because $\text{COG} - \text{HDG}$ then measures the current, which is the §6-L4 input.

Non-tidal water is not merely where COG is better; it is the only place where the compass offset is identifiable at all.

18. Time-resolved TWD needs a quality gate, not a better fit

A naive 45-minute rolling re-fit failed badly — median error **75°** against the model, with windows returning 5° and 118°. The failing windows were **two-boat speed-test runs held on a single tack**, which genuinely cannot yield an absolute TWD: there is no opposite tack to bisect against.

Adding a gate — both tacks present upwind for ≥ 2 min, symmetry score ≥ 0.35 — and pooling both boats:

window (local)	pooled TWD	Open-Meteo
13:15	270°	273°
13:30	268°	273°
13:45	266°	273°
14:00	263°	284°
12:30 / 12:45 / 13:00 / 14:15	unmeasured	—

Median error 7°, max 21°, with four windows correctly refusing to answer.

Two things this confirms empirically rather than by argument:

1. **The "mark unmeasured rather than invent" principle (§8) is not a nicety.** The difference between a 75° median error and a 7° one is entirely the gate.
2. **Pooling rescues individual failures.** At 13:45 boat A alone returned 140°; pooled with boat B the answer was 266°. This is the §5-L5 fleet multiplier observed directly, on two boats rather than five.

It also confirms the architecture: **a windowed re-fit is the wrong shape.** Absolute anchors where the geometry supports them, relative tracking (L3) between, exactly as §6 proposed.

19. Agreement with the model, and a corrected claim

COG-derived session TWD **265°** against Open-Meteo's **262–273°** for 11:00–13:00 local. Agreement within a few degrees — and *expected*, because a COG-derived estimate is **ground wind**, the same reference forecasts are issued in (§11).

Correction to an intermediate conclusion: at one point the heading-derived figure of ~256° suggested the model was locally wrong by ~20°. It was not. That gap was the compass bias. The model was right, and the mutual agreement between track and model is now a point of confidence rather than a discrepancy to explain.

20. What Part 4 changes

- §2d, §6-L1, §6-L4, §9.7 — corrected in place; COG is the wind channel.
- **New requirement:** per-device compass offset, solved per boat per session from the crab decomposition, stored on the boat/tracker and carried forward.
- **New requirement:** the estimator must have a quality gate and an **unmeasured** state before it is shown to anyone.
- **Validated:** the fleet multiplier, the ground-wind framing, and the segment-then-fit approach.
- **Still unvalidated:** everything about current (L4), because this venue has none — and TWS inference, which was not attempted.